Natural Language Processing 2

Evaluation Metrics

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Evaluation Metrics

Evaluation metrics help us measure how well a model performs.

There are two main types of evaluation metrics:

- **Training-based metrics:** These are used during model training. For text generation, the model predicts the next token and assigns a probability to each possible token. The predicted probabilities are compared with the correct target tokens using cross-entropy loss.
- **Reference-based metrics:** These are used after generation. The model output is compared with a human-written reference text. This is common in tasks such as translation and summarization.

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Log-Likelihood

Idea: Measure how probable the real sequence of words is according to the model.

Definition:

$$\log P(X) = \sum_{i=1}^{N} \log P(w_i \mid w_1, \ldots, w_{i-1})$$

- w_i: the i-th word in the sentence.
- $P(w_i \mid w_1, \ldots, w_{i-1})$: model's predicted probability for the real next word.
- A higher log-likelihood means the model fits real text better.

Example:

- Sentence: "The student passed the exam."
- The model computes probabilities:

$$P(\text{student}|\text{The}),\ P(\text{passed}|\text{The student}),\ P(\text{exam}|\text{The student passed the})$$

- Log-likelihood is the sum of these log probabilities.

We do not generate words — we measure how likely the real words were under the model.

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Entropy

Idea: Entropy measures how uncertain a model is when predicting the next word.

Formula:

$$H = -\frac{1}{N}\sum_{i=1}^{N} \log P(w_i \mid w_1, \ldots, w_{i-1})$$

- It's the **average negative log-likelihood per word**.
- Lower entropy means the model is more confident and consistent.

Intuition:

- A model with low entropy "knows" the language well.
- A model with high entropy is uncertain about what word comes next.

Entropy captures the model's uncertainty — the lower, the better.

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Perplexity

Definition:

$$\text{Perplexity} = e^{H} = e^{-\frac{1}{N}\sum_{i=1}^{N}\log P(w_i|w_{<i})}$$

- It's the exponential of the average negative log-likelihood.
- Lower perplexity = better model.

Intuition:

- Perplexity measures how many choices, on average, the model is "confused" among.
- A model with perplexity = 50 acts like it's choosing from 50 equally likely words each time.

Example:

$$\text{If } H = 3.0, \quad PPL = e^{3} \approx 20.1$$

The model is about as uncertain as guessing among 20 likely words each step.

Perplexity is the standard metric because it's interpretable and positive.

Relationship

$$\text{Log-Likelihood} \rightarrow \text{Entropy} \rightarrow \text{Perplexity}$$

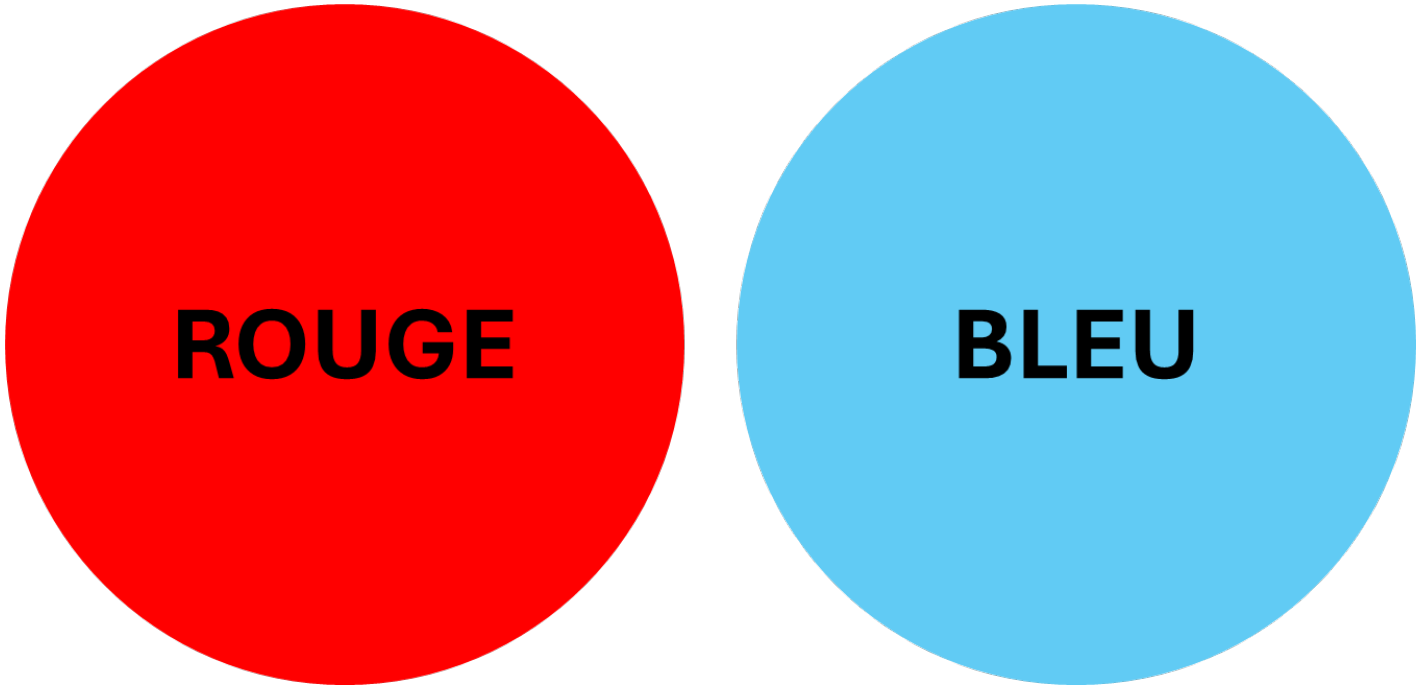
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ROUGE and BLEU



Unlike previous metrics, which measure how well a model predicts real text, BLEU and ROUGE evaluate how close **generated text** is to **human-written references**.

Evaluation Metrics

- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)**
  - Measures overlap of **n-grams**, word sequences, or word pairs between candidate and reference.
  - Commonly used for **text summarization**.
  - Variants: ROUGE-1 (unigrams), ROUGE-2 (bigrams), ROUGE-L (longest common subsequence).
- **BLEU (Bilingual Evaluation Understudy)**
  - Precision-based metric measuring **n-gram overlap**.
  - Widely used in **machine translation**.

Definition of Unigram, Bigram, N-gram

Example: I love pizza.

- **Unigram**: A single token (word).
  - Example: { “I”, “love”, “pizza” }
- **Bigram**: A sequence of 2 consecutive tokens.
  - Example: { “I love”, “love pizza” }
- **N-gram**: A sequence of n consecutive tokens.
  - Example (trigrams, $n = 3$): { “I love pizza” }

Another Example — Unigram, Bigram, N-gram

Example sentence: *"The cat sat on the mat."*

- **Unigrams** (single words): { "The", "cat", "sat", "on", "the", "mat" }
- **Bigrams** (two consecutive words): { "The cat", "cat sat", "sat on", "on the", "the mat" }
- **Trigrams** ($n = 3$ consecutive words): { "The cat sat", "cat sat on", "sat on the", "on the mat" }
- **General N-grams**: Any sequence of n consecutive tokens.

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ROUGE-1 Example

Reference (human): "She is happy"
Candidate (LLM output): "She is very happy"

Unigrams:

- Reference (3): {She, is, happy}
- Candidate (4): {She, is, very, happy}
- Matching unigrams: multiset intersection = 3 {She, is, happy}

Formulas:

$$\text{Recall} = \frac{\text{matching unigrams}}{\text{reference length}} = \frac{3}{3} = 1.00$$

$$\text{Precision} = \frac{\text{matching unigrams}}{\text{candidate length}} = \frac{3}{4} = 0.75$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} = \frac{2 \cdot 1.00 \cdot 0.75}{1.00 + 0.75} \approx 0.86$$

Recall and Precision Formula

$$\text{Recall} = \frac{\text{matching n-grams}}{\text{number of n-grams in reference}}$$

$$\text{Precision} = \frac{\text{matching n-grams}}{\text{number of n-grams in candidate}}$$

ROUGE-1 Limitation

Reference (human): "She is happy"
Candidate (LLM output): "She is not happy"

Unigrams:

- Reference (3): {She, is, happy}
- Candidate (4): {She, is, not, happy}
- Matching unigrams: multiset intersection = 3 {She, is, happy}

Formulas:

$$\text{Recall} = \frac{\text{matching unigrams}}{\text{reference length}} = \frac{3}{3} = 1.00$$

$$\text{Precision} = \frac{\text{matching unigrams}}{\text{candidate length}} = \frac{3}{4} = 0.75$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} = \frac{2 \cdot 1.00 \cdot 0.75}{1.00 + 0.75} \approx 0.86$$

Same Result!

ROUGE-1 Limitation

Observation: ROUGE-1 only measures unigram overlap.

Even if two sentences have **opposite meanings**, they may still share most words. This leads to **high recall, precision, and F1 values**, even though the semantics are wrong.

Example:

- Reference: "She is happy"
- Candidate: "She is not happy"

ROUGE-1 gives high scores here, but the meaning is the opposite.

Another ROUGE-1 Limitation

Reference (human): "The dog chased the cat"

Candidate (LLM output): "dog dog dog dog"

Naive, incorrect precision without clipping:
Unigram matches = 4
Unigrams in candidate = 4

$$\text{Precision} = \frac{4}{4} = 1.0 \quad :($$

Modified precision with clipping:

$$\begin{aligned}\text{clipped match count for "dog"} &= \min(\text{count}_{cand}(\text{dog}), \text{count}_{ref}(\text{dog})) \\ &= \min(4, 1) = 1\end{aligned}$$

Unigrams in candidate = 4

$$\text{Precision} = \frac{1}{4} = 0.25 \quad :)$$

Takeaway: Clipping prevents repeated words from being over-counted.

Does Clipping really solve the problem?

Reference (human): "The dog chased the cat"
Generated output: "cat chased the dog"

Modified precision:
Unigram matches (clipped) = 4, because every candidate word appears in the reference at least once.
Unigrams in output = 4

$$\text{Precision} = \frac{4}{4} = 1.0 \quad :($$

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ROUGE-2 Example

Reference (human): "She is happy"
Candidate (LLM output): "She is not happy"

Bigrams:

- Reference (2): {She is, is happy}
- Candidate (3): {She is, is not, not happy}
- Matching bigrams: multiset intersection = 1 {She is}

Formulas:

$$\text{Recall} = \frac{\text{matching bigrams}}{\text{reference bigrams}} = \frac{1}{2} = 0.50$$

$$\text{Precision} = \frac{\text{matching bigrams}}{\text{candidate bigrams}} = \frac{1}{3} \approx 0.33$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} = \frac{2 \cdot 0.50 \cdot 0.33}{0.50 + 0.33} \approx 0.40$$

ROUGE-2 Another Example

Reference (human): "The cat sat on the mat"
Candidate (LLM output): "The cat sat on the mat happily"

Bigrams:

- Reference (5): {The cat, cat sat, sat on, on the, the mat}
- Candidate (6): {The cat, cat sat, sat on, on the, the mat, mat happily}
- Matching bigrams: multiset intersection = 5 {The cat, cat sat, sat on, on the, the mat}

Formulas:

$$\text{Recall} = \frac{5}{5} = 1.00$$

$$\text{Precision} = \frac{5}{6} \approx 0.83$$

$$F_1 \approx 0.91$$

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ROUGE-L (Longest Common Subsequence)

Definition: ROUGE-L measures overlap via the Longest Common Subsequence (LCS), rewarding matches that preserve order (not necessarily contiguous).

Reference (human): "The cat ran very quickly through the garden and sat on the mat"

Candidate (LLM output): "The cat sat on the mat"

Longest Common Subsequence: [The, cat, sat, on, the, mat] (length = 6)

Lengths: $|\text{ref}| = 13,\ |\text{cand}| = 6$

Formulas:

$$\text{Recall} = \frac{6}{13} \approx 0.46$$

$$\text{Precision} = \frac{6}{6} = 1.00$$

$$F_1 = \frac{2 \cdot 1.00 \cdot 0.46}{1.00 + 0.46} \approx 0.63$$

ROUGE-L Another Example

Reference (human): "The cat sat on the mat"
Candidate (LLM output): "The cat slept on the mat"

Longest Common Subsequence: [The, cat, on, the, mat] (length = 5)
Lengths: $|\text{ref}| = 6,\ |\text{cand}| = 6$

Formulas:

$$\text{Recall} = \frac{5}{6} \approx 0.83$$

$$\text{Precision} = \frac{5}{6} \approx 0.83$$

$$F_1 = \frac{2 \cdot 0.83 \cdot 0.83}{0.83 + 0.83} \approx 0.83$$

Practice: ROUGE-L (Longest Common Subsequence)

Reference (human): "The dog chased the cat"
Candidate (LLM output): "The cat chased the dog"

Solution

Longest Common Subsequence: [The, chased, the] (length = 3)
Lengths: $|\text{ref}| = 5$, $|\text{cand}| = 5$

$$\text{Recall} = \frac{3}{5} = 0.60$$

$$\text{Precision} = \frac{3}{5} = 0.60$$

$$F_1 = \frac{2 \cdot 0.60 \cdot 0.60}{0.60 + 0.60} = 0.60$$

This is a limitation because ROUGE-L still gives a relatively high score even though the roles of the dog and the cat are reversed, which changes the meaning.

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ROUGE-L vs. ROUGE-Lsum

Both metrics use the **Longest Common Subsequence (LCS)** to measure word overlap while preserving word order.

ROUGE-L:

- Treats the entire text as one sequence.
- Computes one LCS between the reference and candidate.
- Commonly used for sentence-level comparison.

ROUGE-Lsum:

- Designed for multi-sentence summaries.
- Uses sentence boundaries.
- Combines sentence-level matches using **union-LCS**.

Key difference:
ROUGE-L evaluates word order across the full text, while ROUGE-Lsum accounts for sentence structure.

Example: Why Sentence Boundaries Matter

Reference:
"The cat slept. The dog barked."

Candidate:
"The dog barked. The cat slept."

Both summaries contain the same information, but the sentence order is reversed.

ROUGE-L:

- Treats both summaries as single sequences.
- The reversed sentence order reduces the LCS.

ROUGE-Lsum:

- Matches sentences separately using union-LCS.
- Can recover all six words despite the reversed sentence order.

Takeaway: ROUGE-Lsum is less sensitive to changes in sentence order than ROUGE-L.

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BLEU Metric

Definition: BLEU measures n-gram *precision* overlap between a candidate and one or more references, using *modified* precision, where each n-gram count is clipped to its maximum frequency in the references.

BLEU also uses a *brevity penalty* to penalize candidates that are too short.

Usual setup:

- Use $n = 1, \ldots, N$, usually with $N = 4$.
- Combine per-n precisions with a *weighted geometric mean*.
- Multiply by the brevity penalty.

$$p_n = \frac{\sum_{g \in \mathcal{G}_n(\text{cand})} \min(\text{count}_{\text{cand}}(g), \text{count}_{\text{ref}}^{\max}(g))}{\sum_{g \in \mathcal{G}_n(\text{cand})} \text{count}_{\text{cand}}(g)}$$

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \qquad \sum_{n=1}^{N} w_n = 1 \ (\text{typically } w_n = \frac{1}{N})$$

BLEU Brevity Penalty

Reference (human): "The cat sat on the mat"

Candidate (LLM output): "The cat"

The candidate matches the beginning of the reference, but it is too short.

Lengths:

$$c_{\text{candidate}} = 2, \qquad r_{\text{reference}} = 6$$

Since $c < r$, BLEU applies a brevity penalty:

$$\text{BP} = e^{1-r/c}$$

$$\text{BP} = e^{1-6/2} = e^{-2} \approx 0.14$$

Takeaway: BLEU penalizes short candidates, even when their precision is high.

Why BLEU Usually Means BLEU-4

When people report **BLEU**, they usually mean one combined score.

This score usually combines modified precision for:

- 1-grams
- 2-grams
- 3-grams
- 4-grams

Each *n*-gram level measures a different kind of overlap:

The final BLEU score combines these using a weighted geometric mean and applies a brevity penalty.

Takeaway:
Unless stated otherwise, **BLEU** usually means **BLEU-4**, a combined score using 1-gram to 4-gram modified precision.

BLEU vs. ROUGE: Simple Difference

ROUGE asks:
How much of the reference did the candidate recover?

So ROUGE is mainly **recall-oriented**.

$$\text{ROUGE Recall} = \frac{\text{matched words from reference}}{\text{words in reference}}$$

It cares about whether the output covers the human answer.

BLEU asks:
How much of the candidate is correct according to the reference?

So BLEU is mainly **precision-oriented**.

$$\text{BLEU Precision} = \frac{\text{matched words from candidate}}{\text{words in candidate}}$$

It cares about whether the generated output avoids adding wrong or extra words.

Important distinction

Important distinction:

- ROUGE as a metric family is usually reported and interpreted as **recall-oriented**.
- BLEU is built around modified **precision** as its main score.

Common use cases:

- ROUGE is commonly used for **summarization**.
- BLEU is commonly used for **machine translation**.

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METEOR Metric

Definition: METEOR evaluates a candidate text by comparing it with a reference text using word matches.

Unlike BLEU and ROUGE, METEOR can consider:

- Exact word matches
- Stem matches, such as “running” and “run”
- Synonym matches, such as “happy” and “glad”. METEOR knows synonyms through a dictionary or lexical database.
- Word order through a fragmentation penalty

Main idea:
A good candidate should match the reference words and keep a similar word order.

METEOR: Precision and Recall

Reference (human): "The cat sat on the mat"

Candidate (LLM output): "The cat is on the mat"

Matched words:

$$\{\text{The, cat, on, the, mat}\}$$

Counts:

$$m = 5, \quad |\text{candidate}| = 6, \quad |\text{reference}| = 6$$

$$P = \frac{m}{|\text{candidate}|} = \frac{5}{6} \approx 0.83$$

$$R = \frac{m}{|\text{reference}|} = \frac{5}{6} \approx 0.83$$

METEOR: F-Mean

METEOR combines precision and recall, but gives more weight to recall.

$$F_{\text{mean}} = \frac{10PR}{R + 9P}$$

Using:

$$P = 0.83, \qquad R = 0.83$$

$$F_{\text{mean}} = \frac{10 \cdot 0.83 \cdot 0.83}{0.83 + 9 \cdot 0.83} = 0.83$$

Takeaway: METEOR cares about both precision and recall, but recall matters more. Since precision gets the larger weight in the denominator, low recall hurts the final score more. If R is low, the denominator changes less, but the numerator PR drops directly.

METEOR: Fragmentation Penalty

METEOR also checks whether matched words appear in a similar order.

Reference: "The cat sat on the mat"

Candidate: "The mat sat on the cat"

Many words match, but the order is wrong.

METEOR applies a penalty when matched words are split into many separated chunks.

$$\text{Penalty} = \gamma \left(\frac{\text{chunks}}{\text{matches}} \right)^{\beta}$$

Final METEOR score:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty})$$

METEOR: Fragmentation Penalty Example

Using the previous example:
Reference: "The cat sat on the mat"
Candidate: "The mat sat on the cat"
All 6 words match, so:

$$\text{matches} = 6$$

But the matched words are not in the same order. They are split into several chunks:

$$\text{chunks} = 4$$

$$[\text{The}], \quad [\text{mat}], \quad [\text{sat, on, the}], \quad [\text{cat}]$$

METEOR uses:

$$\text{Penalty} = \gamma \left(\frac{\text{chunks}}{\text{matches}} \right)^{\beta}$$

Using common values $\gamma = 0.5$ and $\beta = 3$:

$$\text{Penalty} = 0.5 \left(\frac{4}{6} \right)^{3} \approx 0.15$$

Takeaway: More chunks means worse word order, so the final METEOR score is reduced.

METEOR Example: Synonym Matching

Reference (human): "She is happy"

Candidate (LLM output): "She is glad"

BLEU and ROUGE may only match:

$$\{\text{She}, \text{is}\}$$

METEOR can also match:

$$\text{happy} \leftrightarrow \text{glad}$$

So METEOR gives a better score when the meaning is similar but the wording is different.

Takeaway: METEOR is more flexible than simple word-overlap metrics.

METEOR Limitation

Reference (human): “She is happy”

Candidate (LLM output): “She is not happy”

Many words still match:

$$\{\text{She}, \text{is}, \text{happy}\}$$

But the meaning is opposite because of “not”.

Limitation: METEOR improves over BLEU and ROUGE, but it still does not fully understand meaning.

BLEU vs. ROUGE vs. METEOR

- **BLEU:** mainly precision-oriented.
- **ROUGE:** mainly recall-oriented.
- **METEOR:** uses both precision and recall, with more weight on recall.

METEOR also improves matching by using:

- Stemming
- Synonyms
- Word-order penalty

Simple summary:
METEOR is usually more flexible than BLEU and ROUGE, but it is still based on surface matching.

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BERTScore Metric

Definition: BERTScore compares a candidate text with a reference text using contextual word embeddings from BERT-like models.

Unlike BLEU, ROUGE, and METEOR, BERTScore does not rely only on exact word overlap.

It can capture semantic similarity between words in context.

Main idea:
Words with similar meanings should receive high similarity scores, even if they are not exactly the same.

BERTScore Example

Reference (human): "She is happy"

Candidate (LLM output): "She is glad"

BLEU and ROUGE mainly match:

$$\{\text{She}, \text{is}\}$$

BERTScore can recognize that:

$$\text{happy} \approx \text{glad}$$

So BERTScore gives a higher score because the meanings are similar.

How BERTScore Works

1. Encode the reference and candidate using a pretrained language model.
2. Get a contextual embedding for each token.
3. Compare each candidate token with each reference token using cosine similarity.
4. Match each token to its most similar token in the other sentence.
5. Compute precision, recall, and F1.

Key idea:
BERTScore compares meanings through embeddings, not just exact words.

BERTScore Precision

Question:
How well are the candidate tokens supported by the reference?

For each candidate token, find the most similar reference token.

$$P_{\text{BERT}} = \frac{1}{|\text{cand}|} \sum_{x_i \in \text{cand}} \max_{y_j \in \text{ref}} \cos(x_i, y_j)$$

Average those maximum similarity scores across all candidate tokens.

Interpretation:
High precision means the candidate does not contain many unsupported words.

BERTScore Recall

Question:
How well does the candidate cover the reference tokens?

For each reference token, find the most similar candidate token.

$$R_{\text{BERT}} = \frac{1}{|\text{ref}|} \sum_{y_j \in \text{ref}} \max_{x_i \in \text{cand}} \cos(y_j, x_i)$$

Average those maximum similarity scores across all candidate tokens.

Interpretation:
High recall means the candidate covers most of the reference meaning.

BERTScore F1

BERTScore combines precision and recall using F1.

$$F1_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

Interpretation:

- High precision: candidate content is supported by the reference.
- High recall: candidate covers the reference content.
- High F1: candidate is both accurate and complete.

BERTScore vs. Word-Overlap Metrics

Reference: "The child purchased a bicycle"

Candidate: "The kid bought a bike"

BLEU and ROUGE may give low scores because many words are different.

BERTScore can detect semantic similarity:

$$\text{child} \approx \text{kid}$$

$$\text{purchased} \approx \text{bought}$$

$$\text{bicycle} \approx \text{bike}$$

Takeaway: BERTScore is better when meaning is preserved but wording changes.

BERTScore Limitation

Reference: “She is happy”

Candidate: “She is not happy”

The sentences share many similar contextual tokens.

But the word “not” changes the meaning.

Limitation:
BERTScore is more semantic than BLEU and ROUGE, but it can still miss important logical changes such as negation.

BLEU vs. ROUGE vs. METEOR vs. BERTScore

- **BLEU:** mainly precision-oriented, based on *n*-gram overlap.
- **ROUGE:** mainly recall-oriented, based on word or sequence overlap.
- **METEOR:** uses precision, recall, stemming, synonyms, and word-order penalty.
- **BERTScore:** uses contextual embeddings to measure semantic similarity.

Simple summary:
BERTScore moves from surface word matching toward meaning-based comparison.

Score Ranges

Most text evaluation scores are reported between 0 and 1.

$$0 \leq \text{score} \leq 1$$

Interpretation:

- 0 means no useful match.
- 1 means a perfect match.

This applies to:

- ROUGE precision, recall, and F1
- BLEU
- METEOR
- BERTScore, in most common reporting settings. BERTScore is usually reported between 0 and 1, but because it is based on cosine similarity between contextual embeddings, the exact range can depend on the model, implementation, and whether the score is rescaled.

Some tools multiply scores by 100.

Congratulations!

Now you know about the evaluation metrics!

